

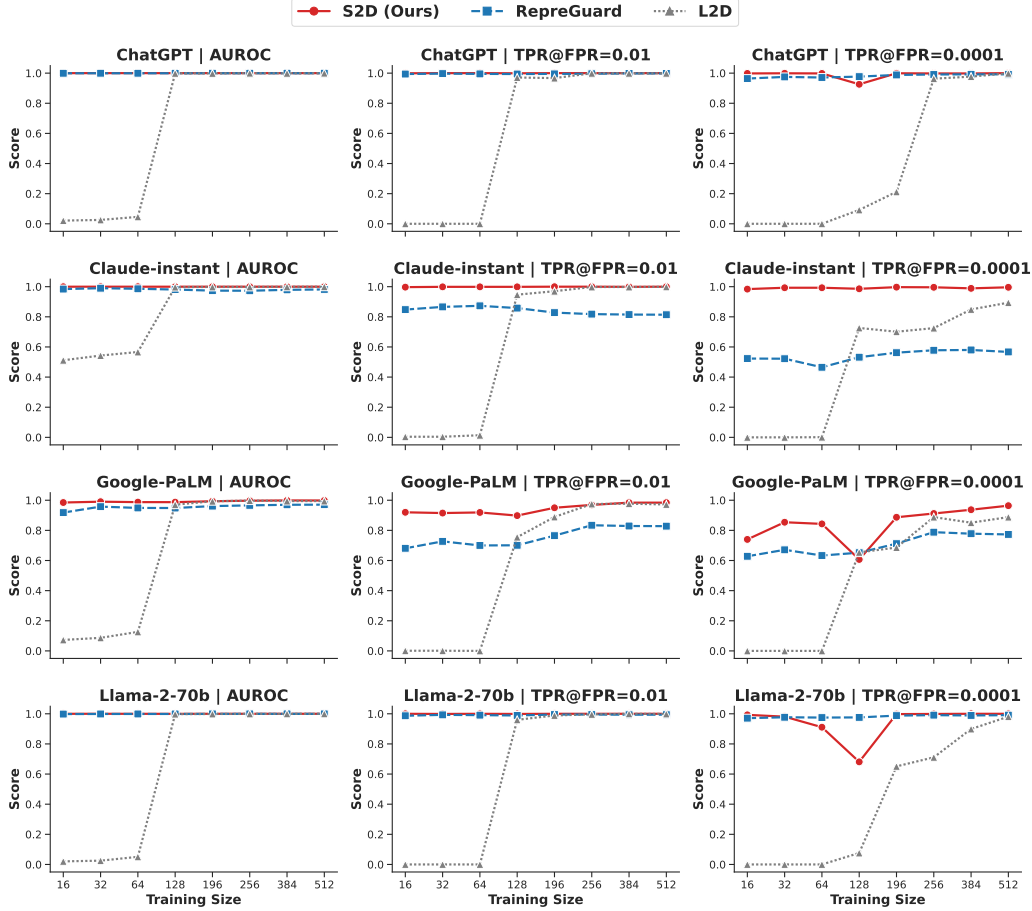


Figure 8: Detection performance across different training set sizes.

Table 10: Steering-layer sweeps across model scales. Here,  $r = \ell/L$  denotes normalized injection depth. TPR@0.01% and overlap are reported at the best validated layer as mean  $\pm$  standard deviation over three seeds.

| Observer         | Swept layers               | High-performing interval           | Best layer $\ell/L$ | TPR@0.01%        | Overlap (%)     |
|------------------|----------------------------|------------------------------------|---------------------|------------------|-----------------|
| Llama-3.2-1B     | 2, 4, 6, 8, 10, 12, 14     | 4–6 ( $r = 0.25\text{--}0.375$ )   | 4/16                | $94.09 \pm 2.94$ | $2.29 \pm 0.10$ |
| Meta-Llama-3-8B  | 4, 8, 12, 16, 20, 24, 28   | 8–16 ( $r = 0.25\text{--}0.50$ )   | 12/32               | $97.14 \pm 0.66$ | $1.92 \pm 0.36$ |
| Meta-Llama-3-70B | 10, 20, 30, 40, 50, 60, 70 | 20–50 ( $r = 0.25\text{--}0.625$ ) | 40/80               | $97.52 \pm 0.54$ | $1.97 \pm 0.20$ |

**Steering Depth Across Model Scales.** We examine how the preferred steering depth changes with model scale by sweeping injection layers on *Llama-3.2-1B*, *Meta-Llama-3-8B*, and *Meta-Llama-3-70B* under the same experimental protocol. Since AUROC is nearly saturated, we report TPR@0.01% and human–LLM score overlap. We define the high-performing interval as the largest contiguous region among the swept layers whose mean TPR@0.01% is within 10 percentage points of the model-specific maximum and whose overlap is at most 2.5%.

In Table 10, across all three scales, strong performance is attained over a range of intermediate depths rather than at a single layer. The best normalized depths shift from 0.25 to 0.375 and 0.50, while the high-performing region broadens from 0.25–0.375 to 0.25–0.625. Thus, the results do not support an exact proportional depth-scaling rule; instead, intermediate layers provide a robust search region whose effective range may broaden as model depth increases.

## F.6 Computational Efficiency

We evaluate the computational efficiency of S2D in terms of both inference latency and training cost, and compare it against representative baselines including L2D, RepreGuard, and Binoculars.

**Experimental Setup.** All efficiency profiling is conducted on a single NVIDIA A100 (80GB) GPU. To ensure a fair evaluation, the training phase for all trainable methods is conducted on a mixed dataset comprising 512 text pairs generated by multiple LLMs, and all inference metrics are evaluated over a correspondingly mixed test set. Our test set has an average length of approximately 267 tokens, with 95% of the samples shorter than 435 tokens.

For all methods, we adopt *meta-llama/Llama-3.1-8B* as the observer/proxy model. For Binoculars, *meta-llama/Llama-3.1-8B* is used as the observer while *meta-llama/Llama-3.1-8B-Instruct* serves as the performer. For L2D, the number of rewrites is set to 4. Unless otherwise specified, we report per-sample inference latency with a batch size of 1. Peak memory usage is measured using `torch.cuda.max_memory_allocated()`. Importantly, for the rewrite-based baseline (L2D), the reported training and inference times strictly *exclude* the time required for the auxiliary LLM to autoregressively generate the rewrites, as these were pre-computed offline.

| Method     | Rewrite? | Efficiency       |                   |                   |                    | Performance (%) |              |
|------------|----------|------------------|-------------------|-------------------|--------------------|-----------------|--------------|
|            |          | Train Time (s) ↓ | Train Cost (GB) ↓ | Infer. Time (s) ↓ | Infer. Cost (GB) ↓ | AUROC ↑         | TPR@1% ↑     |
| Binoculars | No       | -                | -                 | 0.50              | 58.0               | 87.70           | 74.70        |
| L2D        | Yes      | 1213.12          | 45.0              | 2.03              | 43.0               | <b>98.96</b>    | 97.60        |
| RepreGuard | No       | 835.62           | 42.0              | 0.32              | <b>38.0</b>        | 98.42           | 81.47        |
| S2D (ours) | No       | <b>759.46</b>    | <b>40.0</b>       | <b>0.30</b>       | 39.0               | 98.90           | <b>97.75</b> |

Table 11: Comparison of computational cost and detection performance evaluated on a mixed test set. *Train Cost* and *Infer Cost* denote the peak memory allocated during the respective phases. Binoculars is a train-free method, hence its training costs are omitted. Note that for L2D, the heavy computational time required to generate rewrites is excluded from the timing metrics.

**Results.** During **inference**, S2D requires only a single forward pass through the frozen observer LLM. Even when L2D is evaluated under the favorable assumption that rewrite generation is excluded, S2D still achieves a **6.7× speedup** (0.30s vs. 2.03s,  $2.03/0.30 \approx 6.7$ ). Compared with Binoculars, S2D substantially reduces inference latency and lowers memory usage by avoiding the need to load two LLMs simultaneously. Compared with RepreGuard, S2D incurs virtually no additional runtime overhead while maintaining comparable memory usage, indicating that steering introduces negligible deployment cost. During **training**, S2D keeps the observer LLM frozen and optimizes only a lightweight steering vector. As a result, it achieves the lowest training time among the evaluated trainable detectors while maintaining low training memory cost. Overall, these results show that S2D provides a scalable and deployable detection framework without sacrificing accuracy.

## F.7 Interpretability Analysis of the Steering Vector

We further examine what the learned steering vector  $\mathbf{v}$  represents and how it affects the observer model. For this analysis, we use the steering vector learned with *Meta-Llama-3-8B* and evaluate it on an independent held-out set of 512 human–LLM text pairs from DetectRL.

**Vocabulary Projection.** Let  $\mathbf{w}_j$  denote the unembedding direction of token  $j$ . We first examine the alignment between  $\mathbf{w}_j$  and  $\mathbf{v}$  using  $\mathbf{w}_j^\top \mathbf{v}$  and  $\cos(\mathbf{w}_j, \mathbf{v})$ . The most aligned tokens are heterogeneous, including multilingual fragments and incomplete subwords, and the maximum absolute cosine similarity is only 0.074. This suggests that  $\mathbf{v}$  does not correspond to a small set of identifiable lexical features. Moreover, since  $\mathbf{v}$  is injected at an intermediate layer and subsequently transformed by the remaining frozen Transformer blocks, direct vocabulary projection provides only a coarse interpretation of its effect.

**Effect on Output Logits.** We next examine the change in output logits induced by steering. Let  $\Delta\ell_j(x) := \ell_j(x; \mathbf{v}) - \ell_j(x; \mathbf{0})$  denote the change in the output logit of token  $j$ . Tokens whose changes differ most between LLM-generated and human-written text include *incumbent*, *subsequently*, *aforementioned*, *plethora*, *concurrently*, and *rationale*, showing some association with formal written expression. However, direct vocabulary alignment poorly predicts these realized changes. The Spearman correlations between  $\cos(\mathbf{w}_j, \mathbf{v})$  and the overall average change  $\mathbb{E}[\Delta\ell_j(X)]$ , and between  $\cos(\mathbf{w}_j, \mathbf{v})$  and the class-conditional difference  $\mathbb{E}[\Delta\ell_j(X) \mid Y = 1] - \mathbb{E}[\Delta\ell_j(X) \mid$

$Y = 0]$ , are 0.061 and  $-0.035$ , respectively. In addition, the corresponding top-50 token sets have no overlap. Thus, the effect of steering cannot be explained well by nearest-token projections alone.

**Effect on the Final Representation.** Finally, we examine how steering changes the representation used by the detector. Recall from (4) that the detection score depends on the projection of  $f_{\theta, \mathbf{v}}(x)$  onto the direction  $\hat{\mu}_1 - \hat{\mu}_0$ . We therefore consider  $\langle \hat{\mu}_1 - \hat{\mu}_0, f_{\theta, \mathbf{v}}(X) - f_{\theta}(X) \rangle$ . Its average value is  $-1.671$  for human-written text and  $0.388$  for LLM-generated text, yielding an LLM-minus-human contrast of  $2.059$ . Hence, although the same vector  $\mathbf{v}$  is injected for every input, its effect after propagation through the remaining Transformer layers is class dependent rather than a uniform shift of the representation space.

Overall, these results suggest that  $\mathbf{v}$  is better viewed as a context-dependent intervention direction, whose effect emerges through the subsequent frozen Transformer layers and is reflected in the final representation used for detection.

## F.8 Additional Experiments

We further evaluate S2D on the recent OpAI-Bench [69] to examine whether the conclusions from DetectRL continue to hold for more recent generators and more challenging human-to-AI revision settings. OpAI-Bench covers four domains, including abstracts, essays, news, and reports, and contains texts produced or revised by *GPT-5.4*, *GPT-5.4-nano*, *Gemini-2.5-Flash*, and *Qwen3-8B*. Our setup uses 512 training pairs and 302 test documents. In addition to the baselines considered in the main experiments, we include several recent training-free detectors: Implicit Reward Model (*IRM*) [71], *Lastde++* [72], *SpecDetect++* [73], and Normalized Temperature Sensitivity (*NTS*) [74].

**Evaluation on recent generators and detectors.** Table 13 compares both training-free and train-based detectors on OpAI-Bench. Each entry reports AUROC / TPR@1% / TPR@0.1%, in percentage points. For training-free methods, no source-generator training is required. For train-based methods, we report results separately for each training generator; the corresponding diagonal entry is in-distribution, while the remaining columns measure cross-generator transfer. The recent training-free methods improve substantially over several classical likelihood-based detectors at low false-positive rates, with *IRM* generally providing the strongest low-FPR performance among this group. Nevertheless, a substantial gap remains between training-free detection and the strongest train-based approaches.

Across the train-based settings, S2D is generally the strongest method or competitive with the strongest baseline, particularly at low FPRs under cross-generator transfer. For example, when trained on *Gemini-2.5-Flash*, S2D achieves TPR@0.1% of 53.65, 74.92, and 89.40 on *GPT-5.4*, *GPT-5.4-nano*, and *Qwen3-8B*, respectively, substantially exceeding the corresponding values of the other train-based detectors. Similar gains are observed for several other training-test combinations. The advantage is not uniform across every metric, as RoBERTa-Large or ReprGuard can match or slightly outperform S2D in a few settings. Overall, these results show that the strong performance of S2D persists when both the generators and competing detectors are updated.

Table 12: Comparison with discriminative probes. Each entry reports AUROC / TPR@1%.

| Revision | Method     | ID                        | OOD                       |
|----------|------------|---------------------------|---------------------------|
| v4       | LogReg     | 0.92 / 0.45               | 0.87 / 0.34               |
|          | Linear SVM | 0.91 / 0.43               | 0.87 / 0.33               |
|          | MLP        | 0.92 / <b>0.50</b>        | 0.88 / 0.37               |
|          | S2D        | <b>0.94</b> / 0.44        | <b>0.90</b> / <b>0.38</b> |
| v6       | LogReg     | 0.98 / 0.84               | <b>0.96</b> / <b>0.71</b> |
|          | Linear SVM | 0.98 / 0.84               | <b>0.96</b> / 0.70        |
|          | MLP        | 0.98 / 0.85               | 0.95 / 0.70               |
|          | S2D        | <b>0.99</b> / <b>0.86</b> | <b>0.96</b> / <b>0.71</b> |
| v8       | LogReg     | <b>0.99</b> / 0.89        | 0.96 / 0.79               |
|          | Linear SVM | <b>0.99</b> / 0.90        | 0.96 / 0.79               |
|          | MLP        | <b>0.99</b> / 0.91        | 0.96 / 0.77               |
|          | S2D        | <b>0.99</b> / <b>0.92</b> | <b>0.97</b> / <b>0.81</b> |

**Comparison with discriminative probes.** We next examine whether the performance of S2D can be reproduced by directly fitting a standard discriminative classifier to the observer representations. We compare against logistic regression, a linear SVM, and a small MLP. All methods use the same frozen Llama-3-8B observer, representation pooling strategy, and training data, while the probe baselines operate on the corresponding pooled representations without steering. We evaluate on the OpAI-Bench revision trajectories, where v4, v6, and v8 correspond to progressively stronger AI revisions of the same human-written documents, with approximately 50%, 75%, and 100% of the sentences edited, respectively. As shown in Table 12, S2D is generally the strongest method or tied